

CSE499A – Section 15 – Group 03

Md. Rahad Hossain, Md. Yousuf, MD. Rokib Hasan Oli

CityPulse: Intelligent Parking & Traffic Control Platform

Weekly Progress Report – Update 4 (Sprint 4 complete)

I. SPRINT OVERVIEW

This week completed Sprint 4. Phase 2 full fine-tuning of the MobileNetV2 classifier is done and the inference pipeline is confirmed working end-to-end. On the backend, parking space registration, geographic hierarchy restructure, and enhanced profile fields were completed. The frontend authentication flow and new UI designs are also done.

II. SPRINT 4: PHASE 2 FINE-TUNING

Phase 2 unlocks all MobileNetV2 layers and trains end-to-end at learning rate 10^{-5} , allowing the backbone to adapt to parking patterns without overwriting pretrained ImageNet features.

TABLE I
PHASE 2 TRAINING RESULTS (VAL ON UFPR05, UNSEEN LOT)

Epoch	Train Loss	Train Acc	Val Loss	Val Acc
1	0.0134	0.9966	0.0560	0.9816
2	0.0074	0.9984	0.0253	0.9915
3	0.0056	0.9987	0.0314	0.9900
4	0.0046	0.9990	0.0285	0.9910
5	0.0040	0.9991	0.0403	0.9870
6	0.0034	0.9992	0.0374	0.9886
7	0.0032	0.9992	0.0868	0.9729
8	0.0027	0.9993	0.0385	0.9897
9	0.0024	0.9993	0.0306	0.9918
10	0.0022	0.9994	0.0378	0.9893

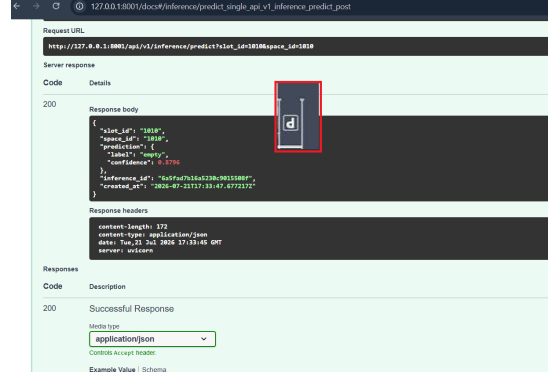
Best: epoch 9, val acc **0.9918** (+1.90 pp over Phase 1). Phase 2 surpasses the best published result on this dataset [3] by 1.17 percentage points.

III. AI INFERENCE PIPELINE: OPERATIONAL

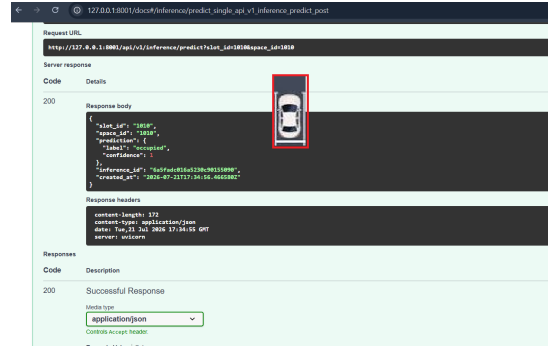
The FastAPI inference service is confirmed working end-to-end. The POST `/api/v1/inference/predict` endpoint accepts a cropped slot image and returns an occupancy prediction with confidence score. Live testing on 21 July 2026 with real slot images confirmed correct classification:

- Occupied slot: label `occupied`, confidence 1.00
- Empty slot: label `empty`, confidence 0.88

Each result carries a unique `inference_id` and UTC timestamp, logged to MongoDB. Single-slot, batch, and history endpoints are all live.



(a) Empty parking slot detected with a confidence score of 0.88.



(b) Occupied parking slot detected with a confidence score of 1.00.

Fig. 1. Live inference results of the proposed parking occupancy detection model.

IV. BACKEND API

A. Parking Spaces Module

A full parking space CRUD module was implemented at `/spaces`. Owners can register a space with name, coordinates, address, capacity, space type, operating hours, amenities, and an optional RTSP camera URL. Up to ten photos may be uploaded per request. Ownership is enforced on all write operations. Soft delete preserves historical data. Seven endpoints are live covering create, read, update, delete, and photo management.

B. Profile and Schema Expansion

User profiles across all three roles were expanded. Driver profiles now store driving licence details following BRTA classification. Owner profiles store business identity documents. The address system was restructured from free-text zone fields to a normalized four-level hierarchy (division → district → thana → area) managed by admin, with all entities referencing a single `area_id` foreign key. Auth DTOs were updated to accept the new fields.

V. FRONTEND

The authentication flow is now complete. Three new pages were implemented this week: login, forgot-password, and reset-password. Together with the signup page from the previous sprint, the full auth entry flow is in place.

Figma designs for the Analytics dashboard and Camera management screens were completed, providing UI references for the authority portal build in Sprint 6.

VI. INFRASTRUCTURE

The backend is now live in a production environment. The database runs on a managed cloud service with connection pooling to handle concurrent requests efficiently. Schema migrations apply automatically on each release so the database stays in sync with the codebase. A build verification step blocks any deployment where the backend fails to compile.

VII. PLAN FOR NEXT WEEK

- Add password reset endpoints to the backend.
- Build the address hierarchy API for cascading dropdown support.
- Begin RTSP ingestion worker in the AI service.
- Analyse fixed-camera CV challenges for Bangladeshi lots.
- Start local parking footage collection for Phase 3 fine-tuning.

REFERENCES

- [1] R. R. Prova *et al.*, “A Real-Time Parking Space Occupancy Detection Using Deep Learning Model,” *IEEE IEMTRONICS*, 2022. DOI: [10.1109/IEMTRONICS55184.2022.9795771](https://doi.org/10.1109/IEMTRONICS55184.2022.9795771)
- [2] G. Amato *et al.*, “Deep learning for decentralized parking lot occupancy detection,” *Expert Systems with Applications*, vol. 72, 2017.
- [3] B. Yuldashev *et al.*, “Parking Lot Occupancy Detection with Improved MobileNetV3,” *Sensors*, vol. 23, no. 17, 2023. DOI: [10.3390/s23177642](https://doi.org/10.3390/s23177642)